Path Traversal in Biological Networks using BFS and DFS (Graded: 20 marks)

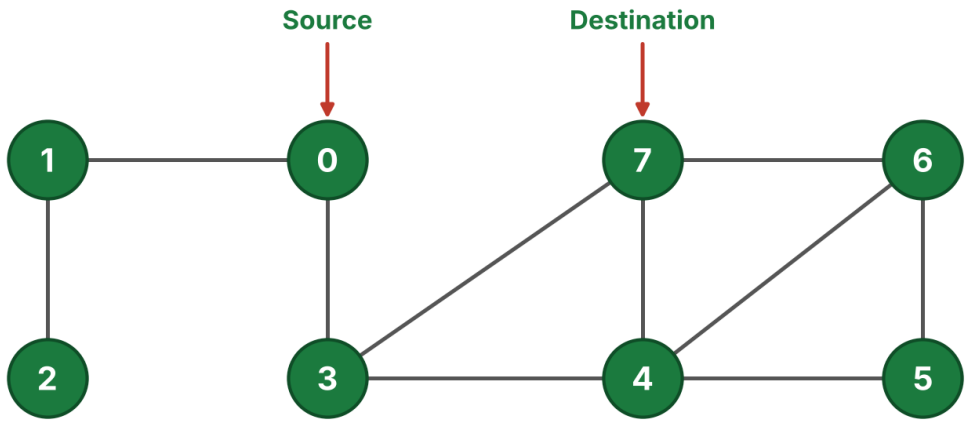
Context

In biological network analysis, such as gene regulatory networks, protein–protein interaction networks or neural connection maps, identifying how nodes (genes, proteins or neurons) are connected helps scientists understand pathways and interactions.

Graph traversal algorithms like Breadth-First Search (BFS) and Depth-First Search (DFS) are key tools for exploring such networks. BFS steps outward one interaction at a time, so it finds the shortest route between two genes. DFS follows one route as deep as it can go before backtracking, so it explores whole pathways.

The network

The undirected graph below represents a small gene interaction network. Each node is a gene (numbered 0 to 7) and each edge is a known interaction between two genes. Gene 0 is the source and gene 7 is the destination.



Edge list (10 interactions): 0–1, 1–2, 0–3, 3–4, 3–7, 4–5, 4–6, 4–7, 5–6, 6–7

Gene	Interacts with
0	1, 3
1	0, 2
2	1
3	0, 4, 7
4	3, 5, 6, 7
5	4, 6
6	4, 5, 7
7	3, 4, 6

Starter code. Copy this adjacency list into your program so that everyone works from the same network.

```
# Gene interaction network (undirected)
graph = {
    0: [1, 3],
    1: [0, 2],
    2: [1],
    3: [0, 4, 7],
    4: [3, 5, 6, 7],
    5: [4, 6],
    6: [4, 5, 7],
    7: [3, 4, 6],
}

SOURCE = 0
DESTINATION = 7
```

Rules

- **Start from gene 0** for both algorithms.
- **Visit neighbors in ascending order.** When a gene has several unvisited neighbors, visit the smallest number first. The lists in the starter code are already sorted.

Tasks

Part A: BFS traversal (5 marks)

Write a function bfs(graph, start) that:

1. Prints the current gene being visited.
2. Prints the list of visited genes after each iteration.
3. Returns the final traversal path as a list (the order in which all genes were visited).

Part B: DFS traversal (5 marks)

Write a function dfs(graph, start) that does the same three things as Part A, using depth-first search.

Part C: Run both algorithms (2 marks)

Run both functions starting from gene 0 and print each final traversal path.

Part D: Path from gene 0 to gene 7 (3 marks)

Extend each traversal to record which gene each new gene was reached from (its parent). Use the parents to rebuild the path from gene 0 to gene 7 found by BFS and the path found by DFS. For each path, state how many interactions (edges) it contains.

Part E: Compare BFS and DFS (2 marks)

How do the traversal paths from BFS and DFS differ? Explain why, referring to the queue used by BFS and the stack (or recursion) used by DFS. Which algorithm gave the shorter path from gene 0 to gene 7?

Part F: REFLECTION (3 marks)

Briefly explain what BFS and DFS might represent in a biological network. For example, consider finding the shortest communication path for a signal between two genes versus exploring a full pathway. Give one example of a biological question each algorithm would suit.

What to submit

Your Python file named `AdmissionNumber_traversal.py`. Write your answers to Part E and Part F in **text cells** at the end of the notebook.

Individual Assignment

Submit before the deadline. Attendance is based on submission.

GOOD DAY!